

Drug Overdoses in the U.S.A

Introduction

Drug overdose mortality in the U.S. has increased substantially since the late 1990s, making it essential to examine differences by age and sex using population-level data. Understanding which groups are most affected is critical for developing targeted interventions and prevention strategies. This study will focus on the total U.S. population from 1999 through 2018 to evaluate disparities in drug overdose death rates using nationally aggregated mortality data. Specifically focusing on how overdose death rates vary across age groups and sex, two key demographic factors widely recognized as important predictors of health outcomes.

The central research question for this study is:

Do drug overdose rates differ significantly by age group and sex in the United States?

In simple terms, this study aims to identify which age groups and which sex experience higher overdose death rates, and whether these differences are statistically meaningful.

We hypothesize that age and sex are significant predictors of drug overdose death rates, with the highest rates occurring among middle-aged adults (35–64) and males, compared to younger (under 35) and older (over 65) populations and females.

Previous research demonstrates that overdose risk is not uniform across the life course or across social categories, which supports the need for analyses that individualize national trends by key demographic variables. We analysed the sample frame by the National Vital Statistics System (NVSS) mortality files, with data representing a census of all death certificates filed in all 50 states and the District of Columbia from 1999 through 2018. By analysing this national data, this study aims to provide a clearer understanding of these demographic groups and to contribute to broader discussion on public health policy.

Literature Review

Age is a central factor that shapes drug overdose risk, as substance abuse patterns and other contributing risk factors differ meaningfully across age groups. In a systematic review of overdose risk among young people, Lyons et al. (2019) found that youth overdose is strongly associated with polysubstance use, injection drug use, psychiatric disorders, and social and environmental instability. These findings suggest that overdose risk among young people is driven by differing ways compared to older adults, which indicates that age is not merely a control variable, but a meaningful dimension of variation. Lyons et al. further emphasize that overdose patterns differ by age group, which justifies the separation of youth and adult populations in overdose research. Supporting the inclusion of age as a categorical explanatory variable informs the hypothesis that overdose death rates are significantly higher among middle-aged adults (from ages 35–54) compared to younger populations under 25 and older populations over the age of 65.

Sex differences in overdose mortality also motivate the inclusion of sex as a key explanatory variable. Ho (2020) examined the long-term trends in drug overdose mortality and found persistent differences between males and females across multiple phases of the overdose epidemic. Although the gender gap narrowed during earlier periods of the overdose epidemic due to prescription opioid use, it widened again after 2010 with the rise of illicit drugs and has remained consistent across racial and ethnic groups. These findings indicate that males are at higher risk of fatal overdose than females over time. This study hypothesizes that males experience significantly higher overdose death rates than females across all drug types in the 1999–2018 period.

Demographic differences also play a significant role in shaping overdose mortality and justify this study's focus on age and sex as key explanatory variables.

Han et al. (2022) analysed national trends in drug overdose deaths and found substantial variation across demographic groups across the population. Although their study emphasises racial and ethnic disparities, its broader findings support the importance of analysing overdose mortality across multiple demographic dimensions such as age and sex and that these demographic characteristics meaningfully influence overdose risk. In this study, we narrow our focus to age and sex to provide a more targeted analysis of two fundamental and consistently measured predictors of overdose mortality while still aiming to align with existing literature emphasizing the importance of demographic variation.

These three studies combined, demonstrate that age, sex, and race are fundamental determinants of overdose mortality in the U.S. Together, they justify the selection of the national population from 1999 through 2018, support the proposed hypotheses, and guide the inclusion of demographic explanatory variables used to explain variation in drug overdose death rates.

Methods

Sample Size

- Our sample size is all available data from the dataset, comprising all annual drug overdose death rates from 1999 through 2018 across all available demographic breakdowns. After cleaning, the dataset contains 476 rows.

Variables

- **Response Variable:** Drug overdose death rate is the quantitative response variable, which measures the number of drug overdose deaths per 100,000 resident population.
- **Explanatory Variables:**
 - **Age** is a categorical explanatory variable with options "Under 15 years," "15–24 years," "25–34 years," "35–44 years," "45–54 years," "55–64 years," "65–74 years," "75–84 years," "85 years and older," used to determine if overdose rates differ by age group.
 - **Sex** is a categorical explanatory variable with options "Male" and "Female," used to determine if overdose rates differ by sex.

Data Source

<https://catalog.data.gov/dataset/drug-overdose-death-rates-by-drug-type-sex-age-race-and-hispanic-origin-united-states-3f72f>

Data Collection Methods

- **Study Type:** Observational study using secondary data, both pre-existing and publicly available.
- **Sub-type:** It is retrospective and longitudinal. The data were collected prospectively by the National Centre for Health Statistics over time (1999–2018), but are being analysed retrospectively by our group after the fact.
- **Sampling Method:** It is an attempted census. The original researchers collected data from all death certificates filed in all 50 states and the District of Columbia through the National Vital Statistics System (NVSS) mortality files. No random sampling was necessary because the dataset represents the entire population of registered drug overdose deaths in the U.S. during the study period.

- **Process:** The National Center for Health Statistics identified all death certificates in the U.S. where drug overdose was listed as the underlying cause of death between 1999 and 2018. Death rates per 100,000 resident population were then calculated and age-adjusted to the year 2000 standard population to allow fair comparisons across demographic groups and over time. The resulting dataset was published publicly on Data.gov, and our group downloaded the CSV file directly from that source.

Gather the data

- **Source:** U.S. Department of Health & Human Services, National Center for Health Statistics. Drug Overdose Death Rates, by Drug Type, Sex, Age, Race, and Hispanic Origin: United States, 1999–2018. Retrieved from Data.gov: <https://catalog.data.gov/dataset/drug-overdose-death-rates-by-drug-type-sex-age-race-and-hispanic-origin-united-states-3f72f>
- **Process:** We used pre-existing, publicly available secondary data published by the National Center for Health Statistics (NCHS). The dataset was downloaded directly from Data.gov as a CSV file. No random sampling was necessary because the dataset contains 6,228 rows, well below the one million row threshold. The data represents a complete record of age-adjusted drug overdose death rates broken down by drug type, sex, age group, race, and Hispanic origin across all 50 states and D.C. from 1999 through 2018.

Clean/Report the Data

- We started with the original dataset of 6,228 observations and filtered it to keep only the rows where STUB_NAME was "Sex and age" since those were the observations relevant to our analysis. Then, we created a new variable called Sex by pulling whether the observation was male or female from the STUB_LABEL column. After that, we removed all rows with missing values using na.omit() so that every observation in the final dataset had complete information for the variables we needed. This helps prevent errors in the analysis and makes the results more reliable. Next, we grouped the original age categories into three broader groups, Under 35, 35 to 64, and 65 plus, to make the analysis simpler and better match our hypothesis. Lastly, we kept only the variables needed for the analysis Year, Sex, AgeGroup, and Estimate. After cleaning, the dataset was reduced from 6,228 observations to 1,651 observations, meaning 4,577 rows were removed during the data cleaning process.

R Code: Data Cleaning

```
library(dplyr)
# Read file
pop <- read.csv(file.choose())

# New data frame that has rows with the STUB_NAME of "Sex and age"
data <- pop[pop$STUB_NAME == "Sex and age",]

# Pull the sex from the STUB_LABEL column
data$Sex = ifelse(grepl("Male", data$STUB_LABEL) == 1, "Male", "Female")
```

```

# Take out NA values
clean_data <- na.omit(data)

# Age grouping based on hypothesis
data_clean <- clean_data %>%
  mutate(
    AgeGroup = case_when(
      AGE %in% c("Under 15 years", "15-24 years", "25-34 years") ~
"Under 35",
      AGE %in% c("35-44 years", "45-54 years", "55-64 years") ~ "35-
64",
      AGE %in% c("65-74 years", "75-84 years", "85 years and over")
~ "65+",
      TRUE ~ NA_character_
    )
  )

# Only keep columns needed for analysis
data_clean <- data_clean %>%
  select(YEAR, Sex, AgeGroup, Estimate)

```

- **Report the data (first 50 rows)**

"YEAR"	"Sex"	"AgeGroup"	"Estimate"
1999	"Male"	"Under 35"	0.1
2000	"Male"	"Under 35"	0.2
2001	"Male"	"Under 35"	0.2
2002	"Male"	"Under 35"	0.2
2003	"Male"	"Under 35"	0.2
2004	"Male"	"Under 35"	0.2
2005	"Male"	"Under 35"	0.2
2006	"Male"	"Under 35"	0.2
2007	"Male"	"Under 35"	0.3
2008	"Male"	"Under 35"	0.2
2009	"Male"	"Under 35"	0.2
2010	"Male"	"Under 35"	0.3
2011	"Male"	"Under 35"	0.2
2012	"Male"	"Under 35"	0.2
2013	"Male"	"Under 35"	0.2
2014	"Male"	"Under 35"	0.2
2015	"Male"	"Under 35"	0.2

2016	"Male"	"Under 35"	0.2
2017	"Male"	"Under 35"	0.2
1999	"Male"	"Under 35"	4.5
2000	"Male"	"Under 35"	5.3
2001	"Male"	"Under 35"	6.1
2002	"Male"	"Under 35"	7.3
2003	"Male"	"Under 35"	8.8
2004	"Male"	"Under 35"	9.6
2005	"Male"	"Under 35"	10
2006	"Male"	"Under 35"	12
2007	"Male"	"Under 35"	12
2008	"Male"	"Under 35"	11.9
2009	"Male"	"Under 35"	11.3
2010	"Male"	"Under 35"	11.6
2011	"Male"	"Under 35"	12.4
2012	"Male"	"Under 35"	11.4
2013	"Male"	"Under 35"	11.7
2014	"Male"	"Under 35"	12.1
2015	"Male"	"Under 35"	13.3
2016	"Male"	"Under 35"	17.5
2017	"Male"	"Under 35"	17.1
1999	"Male"	"Under 35"	11.5
2000	"Male"	"Under 35"	11.3
2001	"Male"	"Under 35"	12
2002	"Male"	"Under 35"	14.7
2003	"Male"	"Under 35"	15.7
2004	"Male"	"Under 35"	16.6
2005	"Male"	"Under 35"	18.7
2006	"Male"	"Under 35"	22.7
2007	"Male"	"Under 35"	23.4
2008	"Male"	"Under 35"	23.6
2009	"Male"	"Under 35"	24
2010	"Male"	"Under 35"	25

- Clean data file: ~/cleaned_drug_overdose.csv
- <https://drive.google.com/file/d/1SOVxywj-dZpogWLYZvjhE0l8ERQikOyp/view?usp=sharing>

Limitations

Before we present the findings of our study, we acknowledge that as a research group, our positionality may affect the study itself. Our group consists of four undergraduate students, three males and one female. Including two Hispanic members and two White members studying in the United States. Our combined academic backgrounds span statistics, psychology, anthropology, data science, accounting, music, and sports management which collectively influenced both our interest in this project and how we approached the research topic. These diverse disciplines have shaped how we think about data, behaviour, and social issues, which all likely guided our focus on demographic patterns in drug overdose death rates such as age, sex, and race.

Our identities and academic experiences influenced our research design and shaped our expectations of the study outcomes. Due to our exposure in broader discussions and coursework around public health, inequality, and the opioid epidemic, we entered this project with prior assumptions that overdose deaths would be higher among middle-aged adults. These assumptions influence our hypotheses and may introduce confirmation bias in how we interpret the results, particularly when findings align with widely discussed narratives or trends of overdose death rates, even when they may seem more nuanced.

Additionally, we do not share the same lived experiences of individuals most affected by drug overdose, which may in turn, limit our ability to fully contextualize the data. Our emphasis on quantitative analysis may overlook the qualitative factors not captured in the dataset. This is likely due to our academic training particularly in statistics and data science that may lead us to prioritize quantitative patterns over lived experiences. By recognizing these biases and assumptions, we can approach our findings with greater transparency and caution.

One limitation of our study is the exclusion of race and ethnicity since we narrowed its focus down to age and sex. Prior research has shown that overdose mortality varies across different racial and ethnic groups, and omitting these groups may overlook important disparities and interacting effects. Thus, this study may not fully capture the complexity of identity and overdose rates, however, this decision was made to maintain a more focused analysis on age and sex. Two consistently measured explanatory variables that allow for clearer statistical comparisons across groups, as some studies may include varying racial categories in original data collection. Future research should incorporate race and ethnicity alongside both age and sex to provide a more nuanced understanding of overdose patterns.

This study suffers from undercover bias. This is because there are people who have died from drug overdoses that haven't had their cause of death properly recorded. For example, if someone overdosed and wasn't admitted to the hospital it would be hard to get that data recorded for a study like this. Non-response bias isn't present in this data because it wasn't a survey that was sent out, it is based on facts and data that was collected from public use mortality files. Similarly, response bias also doesn't play a factor for the same reasons, someone can't lie about their response because this data wasn't collected via survey.

Our study could suffer from lurking variables. There are many different reasons that someone might overdose on drugs other than just their sex and age. For example, the environment in which someone lives could have an impact on access to drugs. People that live in communities that have higher access to drugs are more likely to overdose. Another lurking variable that could be present in our study is whether the people who overdosed had adequate access to healthcare and resources to combat addiction. If people didn't have access to the proper health or resources, they may have been more likely to overdose. For example, if someone was battling addiction and didn't have the ability to go to rehab, they are probably more likely to overdose. In comparison, people that do have that ability are much more likely to move past their addiction. The location that someone lives might impact our location because we would wrongly be looking at race as a determinant rather than the location that they live in. We might conclude that a certain age group and sex has a higher overdose rate, when it is affected by their location. Similarly, we might conclude that age and sex played a factor in someone's overdose when, in fact, they were struggling with addiction and didn't have the proper resources to fight back against it.

Findings

Exploratory Data Analysis ***deaths per 100,000**

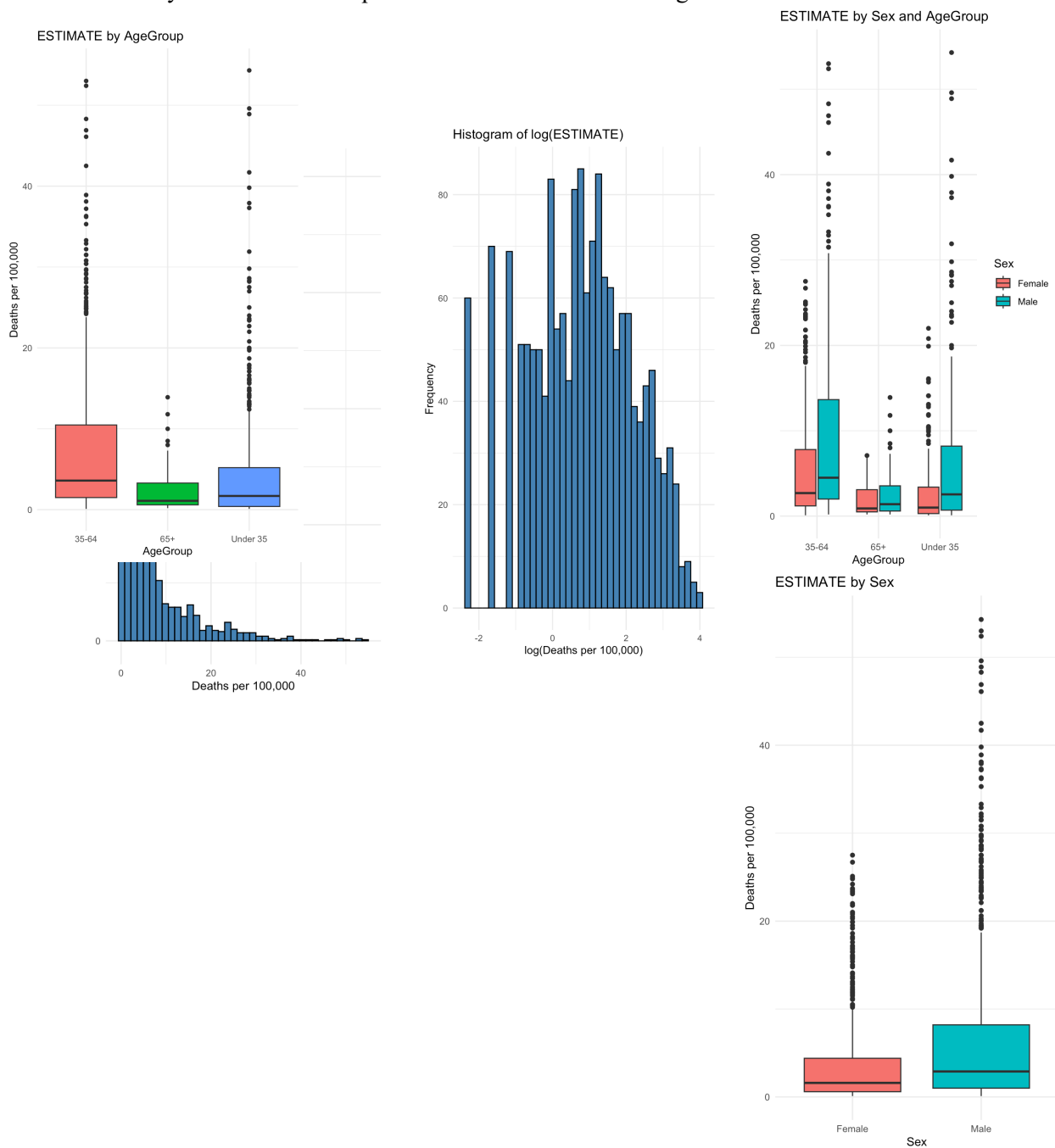
Summary statistics by Sex				
Sex	n	Estimate*	Median	SD
Female	827	3.68	1.6	4.95
Male	824	6.73	2.9	9.14

Summary statistics by AgeGroup				
AgeGroup	n	Estimate*	Median	SD
Under 35	565	4.47	1.7	7.15
35–64	710	7.46	3.6	8.8
65+	376	2.03	1.1	1.95

Summary statistics by Sex × AgeGroup				
Sex	AgeGroup	n	Estimate*	SD
Female	Under 35	275	2.61	3.74
Female	35–64	351	5.61	6.21
Female	65+	201	1.76	1.6
Male	Under 35	290	6.23	8.95
Male	35–64	359	9.27	10.44
Male	65+	175	2.35	2.26

- Males have a higher estimate rate than females across all age groups. The 35–64 age group has the highest mean rate. The response is heavily right-skewed on the raw scale and approximately symmetric after a log

transformation. Within-group standard deviations are very uneven across groups, suggesting a likely violation of the equal variance condition in the regression model.



Correlation

	SexNum	AgeNum	Estimate
SexNum	1	-0.03355752	0.20375047
AgeNum	-0.03355752	1	-0.08393731
Estimate	0.20375047	-0.08393731	1

- Pairwise correlations confirm that Sex is more strongly associated with overdose death rate ($r = 0.204$) than AgeGroup ($r = -0.084$), and the two predictors are nearly uncorrelated with each other ($r = -0.034$), indicating no multicollinearity concern in the model.

Statistical Inference:

Methods: Multiple linear regression with two categorical predictors. Tests used are the overall ANOVA F test, individual t tests for each coefficient, and partial F tests for cross-validation. 95% confidence intervals are reported for each coefficient.

Significance level: $\alpha = 0.05$.

Population model:

$$\mu_y = \beta_0 + \beta_1 \cdot \text{SexMale} + \beta_2 \cdot \text{AgeGroup65+} + \beta_3 \cdot \text{AgeGroupUnder35}$$

Parameter definitions:

- y: Estimate, drug overdose deaths per 100,000, age-adjusted
- β_0 : mean log rate for the reference group (Female, 35–64)
- β_1 : mean difference for Male vs Female, holding AgeGroup constant
- β_2 : mean difference for 65+ vs 35–64, holding Sex constant
- β_3 : mean difference for Under 35 vs 35–64, holding Sex constant

Hypotheses:

Overall ANOVA F test:

- $H_0: \beta_1 = \beta_2 = \beta_3 = 0$
- H_a : at least one $\beta_j \neq 0$

Individual t-tests:

- $H_0: \beta_1 = 0$ vs $H_a: \beta_1 \neq 0$
- $H_0: \beta_2 = 0$ vs $H_a: \beta_2 \neq 0$
- $H_0: \beta_3 = 0$ vs $H_a: \beta_3 \neq 0$

Partial F tests:

- H_0 : reduced model is good enough
- H_a : full model is needed

Initial Model

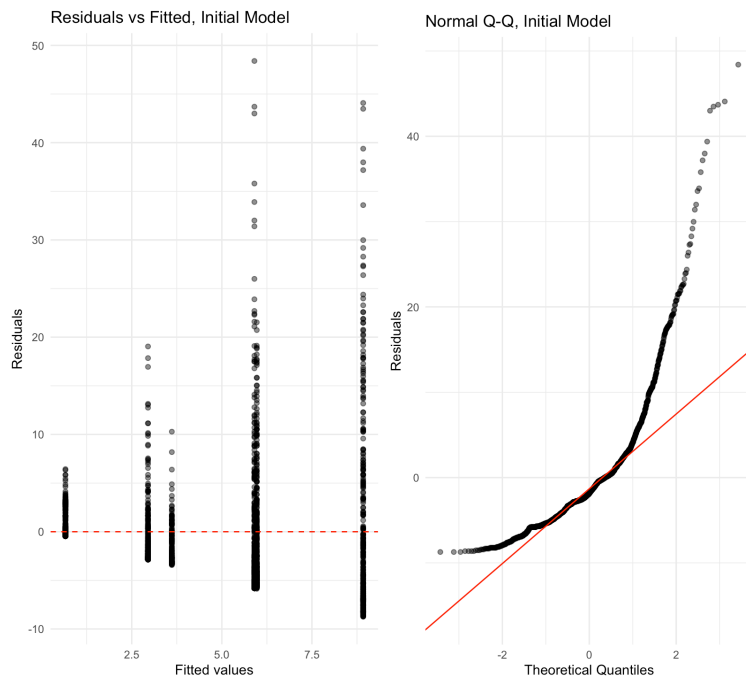
Output: `lm(Estimate ~ Sex + AgeGroup)`

Coefficient	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	5.9698	0.3170	18.833	< 2e-16
SexMale	2.9520	0.3466	8.516	< 2e-16
AgeGroup65+	−5.3089	0.4491	−11.822	< 2e-16
AgeGroupUnder 35	−3.0182	0.3968	−7.607	4.68e-14

Residual standard error: 7.037 on 1,647 df. Multiple R²: 0.1216. Adjusted R²: 0.1200. F-statistic: 76.03 on (3, 1647), p-value < 2.2e-16.

Conditions for Initial Model:

- **Linearity: Yes**, both predictors are categorical, so the model estimates group means rather than slopes.
- **Independence: No**, multiple observations within the same demographic cell come from different drug subcategories, and consecutive years within the same cell are serially correlated. This is a preliminary study and the independence condition is not met for multiple linear regression. However, we are going to continue onward as if the condition were met.
- **Normality: No**, the Q-Q plot curls sharply upward at the right tail, indicating heavy right skew in the residuals.
- **Equal Variance: No**, the Residuals vs Fitted plot shows a textbook fan shape.



Model Modification

- **Problem identified:** Fan shape in Residuals vs Fitted (Equal Variance violated). Upward curl in Q-Q plot (Normality violated).
- **Modification applied:** A log transformation of the response is the prescribed fix for unequal variance. All Estimate values are strictly positive, so $\log(\text{Estimate})$ is well defined.
- **Revised model:** $\log(\mu_y) = \beta_0 + \beta_1 \cdot \text{SexMale} + \beta_2 \cdot \text{AgeGroup65+} + \beta_3 \cdot \text{AgeGroupUnder35}$

Final Model

Output: $\text{lm}(\log(\text{Estimate}) \sim \text{Sex} + \text{AgeGroup})$

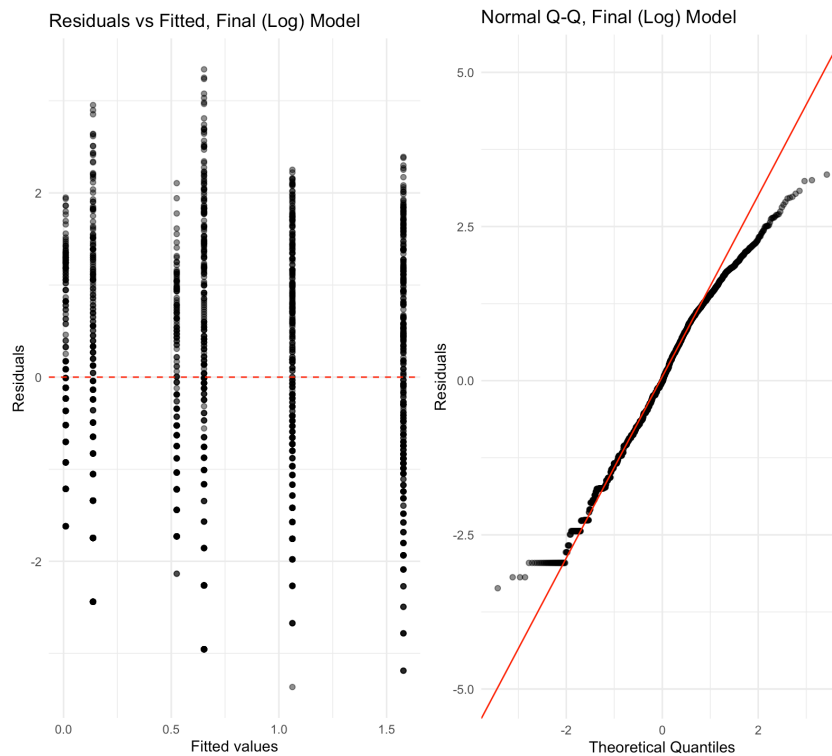
Coefficient	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	1.06260	0.05942	17.882	< 2e-16
SexMale	0.51559	0.06498	7.934	3.88e-15
AgeGroup65+	-1.05275	0.08419	-12.505	< 2e-16
AgeGroupUnder 35	-0.92584	0.07438	-12.448	< 2e-16

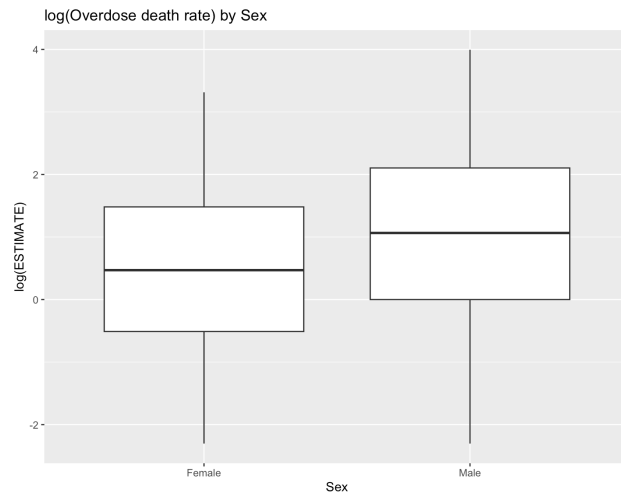
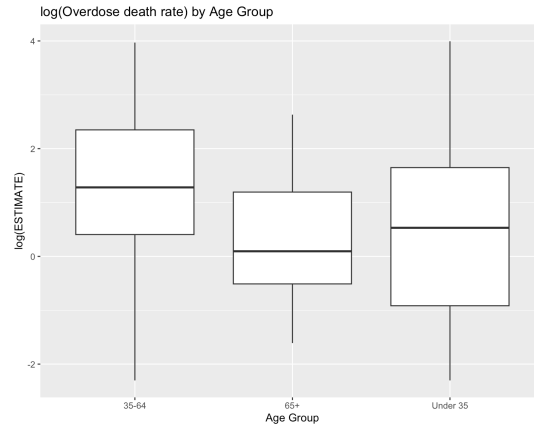
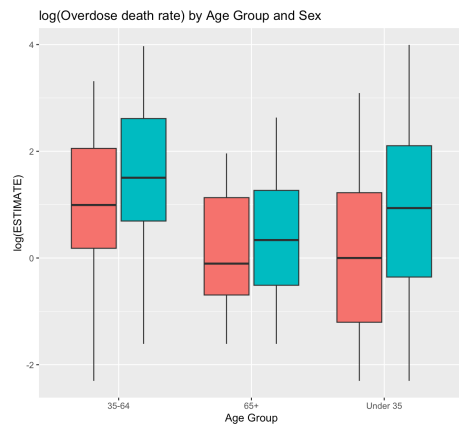
Residual standard error: 1.319 on 1,647 df. Multiple R²: 0.1499. Adjusted R²: 0.1483. F-statistic: 96.79 on (3, 1647), p-value < 2.2e-16.

Conditions for Final Model:

- **Linearity: Yes**, same reasoning as before.
- **Independence: No**, still not met; proceeding as stated above.
- **Normality: Reasonably yes**, the Q-Q plot tracks the 45° reference line closely with only mild deviation at the tails.
- **Equal Variance: Reasonably yes**, the fan shape is gone and residual spreads are uniform across groups.

Adjusted R^2 improved from 0.1200 (initial) to 0.1483 (final), further supporting the log transformation.





Partial F Tests

Drop Sex (q = 1):

Model 1: $\log() \sim \text{AgeGroup}$

Model 2: $\log() \sim \text{Sex} + \text{AgeGroup}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	1648	2976.1				
2	1647	2866.6	1	109.57	62.953	3.878e-15

TS = 62.953, df = (1, 1647), p-value = 3.88e-15. Reject H_0 , Sex is retained.

Drop AgeGroup (q = 2):

Model 1: $\log(\text{Estimate}) \sim \text{Sex}$

Model 2: $\log(\text{Estimate}) \sim \text{Sex} + \text{AgeGroup}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	1649	3256.0				
2	1647	2866.6	2	389.47	111.89	< 2.2e-16

TS = 111.89, df = (2, 1647), p-value $\approx 2.74\text{e-}46$. Reject H_0 , AgeGroup is retained.

R² comparison across models:

Model	R ²	Adjusted R ²
Full: $\log(\text{Estimate}) \sim \text{Sex} + \text{AgeGroup}$	0.1499	0.1483

Reduced: $\log(\text{Estimate}) \sim \text{AgeGroup}$ only	0.1174	0.1163
Reduced: $\log(\text{Estimate}) \sim \text{Sex}$ only	0.0344	0.0338
Raw: $\text{Estimate} \sim \text{Sex} + \text{AgeGroup}$	0.1216	0.1200

95% Confidence Intervals

Log scale:

Coefficient	2.5%	97.5%
(Intercept)	0.9461	1.1792
SexMale	0.3881	0.6430
AgeGroup65+	-1.2179	-0.8876
AgeGroupUnder 35	-1.0717	-0.7800

Because the response is logged, exponentiating the CI gives a rate multiplier. For example, SexMale $\exp(\text{CI}) = (1.47, 1.90)$ means the male rate is $1.47\times$ to $1.90\times$ the female rate.

Back-transformed rate multipliers:

Coefficient	$\exp(2.5\%)$	$\exp(97.5\%)$	Point estimate
SexMale	1.474	1.902	1.675
AgeGroup65+	0.296	0.412	0.349
AgeGroupUnder 35	0.342	0.458	0.396

Interpretations:

- **Sex, Male vs Female:** The 95% CI for β_1 is (0.388, 0.643), which does not contain zero, so Sex is a significant predictor. On the raw rate scale, the male overdose death rate is between 1.47 and 1.90 times the female rate.
- **AgeGroup 65+ vs 35–64:** The 95% CI for β_2 is (-1.218, -0.888), which does not contain zero. The 65+ overdose rate is between 0.30 and 0.41 times the 35–64 rate, roughly 59% to 70% lower.
- **AgeGroup Under 35 vs 35–64:** The 95% CI for β_3 is (-1.072, -0.780), which does not contain zero. The Under 35 overdose rate is between 0.34 and 0.46 times the 35–64 rate, roughly 54% to 66% lower.

Statistical Conclusions:

- **Overall F test:** $TS = 96.79$, $p < 2.2e-16$. There is sufficient evidence at the $\alpha = 0.05$ level to conclude at least one of Sex or AgeGroup has a significant linear relationship with log drug overdose death rate.
- **Sex t test:** $TS = 7.934$, $p = 3.88e-15$. There is sufficient evidence at the $\alpha = 0.05$ level to conclude Sex has a significant linear relationship with log overdose death rate when AgeGroup is held constant.
- **AgeGroup 65+ vs 35–64 t test:** $TS = -12.505$, $p < 2.2e-16$. There is sufficient evidence at the $\alpha = 0.05$ level to conclude the 65+ group has a significantly lower rate than the 35–64 reference group when Sex is held constant.
- **AgeGroup Under 35 vs 35–64 t test:** $TS = -12.448$, $p < 2.2e-16$. There is sufficient evidence at the $\alpha = 0.05$ level to conclude the Under 35 group has a significantly lower rate than the 35–64 reference group when Sex is held constant.
- **Partial F tests:** There is sufficient evidence at the $\alpha = 0.05$ level to conclude the full model is significantly better than either reduced model.

These results support the hypothesis. Age and sex are each statistically significant predictors of drug overdose death rate, with the highest rates occurring among middle-aged adults (35–64) and among males.

Conclusion

In conclusion, we can make the practical inference that middle-aged adults (ages 35–64) and males experienced drug overdose deaths at much higher rates than other demographic groups in the population. Males overdosed at a rate 1.68 times higher than females, and adults ages 35–64 were about 3.5 times more likely to overdose than the other age groups. Our research question asked, “Do drug overdose rates differ significantly by age group and sex in the United States?” Based on our analysis, the answer is yes. Age group and sex can be considered significant predictors of drug overdose deaths because the p-values for both variables were less than .001, making them statistically significant. The data supports our hypothesis that middle-aged adults and males were more likely to overdose.

We can connect these findings back to prior literature. First, Ho found that males were more likely to overdose, which was supported by our results. Next, Lyons et al. discussed the risk of overdosing among younger people and argued that overdose risk in younger populations is often shaped by pathways such as polysubstance abuse. This means that young people who abuse more than one drug are more likely to overdose. Although our results showed the highest overdose rates among middle-aged adults, these prior studies help explain risks that exist across multiple age groups.

This study also has methodological limitations. It likely suffers from undercoverage bias because some overdose deaths may not have been properly recorded. For example, if a person overdosed and was never admitted to a hospital, it may be difficult for that death to be included in public records. Nonresponse bias is not present because this was not a survey. Instead, the dataset was based on public mortality records. Similarly, response bias is not a factor because participants were not self-reporting information, so there was no opportunity for false responses.

Based on these findings, several policy and practice recommendations can be made. Prevention efforts should prioritize groups that had the highest risk, especially middle-aged adults and males. Creating treatment and outreach programs separated by sex may also be beneficial, since males and females may respond differently to treatment approaches. In addition, improving data collection

and reducing undercoverage in future studies would strengthen the accuracy of overdose research and help public health officials make better decisions.

For future research, it would be beneficial to create a new study that includes the most recent eight years of data. Overdose trends may have changed, and a different age group may now be at higher risk. It would also be valuable to examine overdose data on a state-by-state basis. This would help identify which states need the most attention and resources in terms of prevention, treatment, and recovery services.

References

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